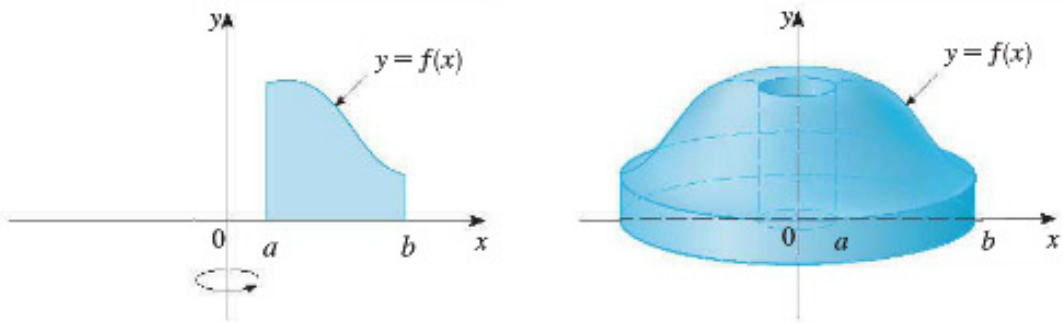
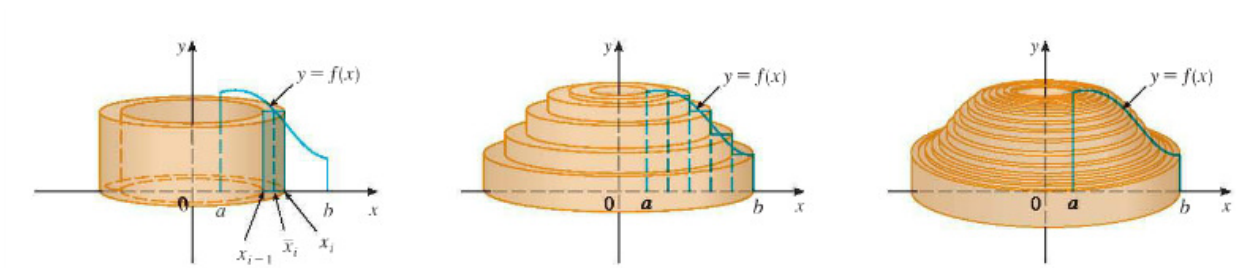
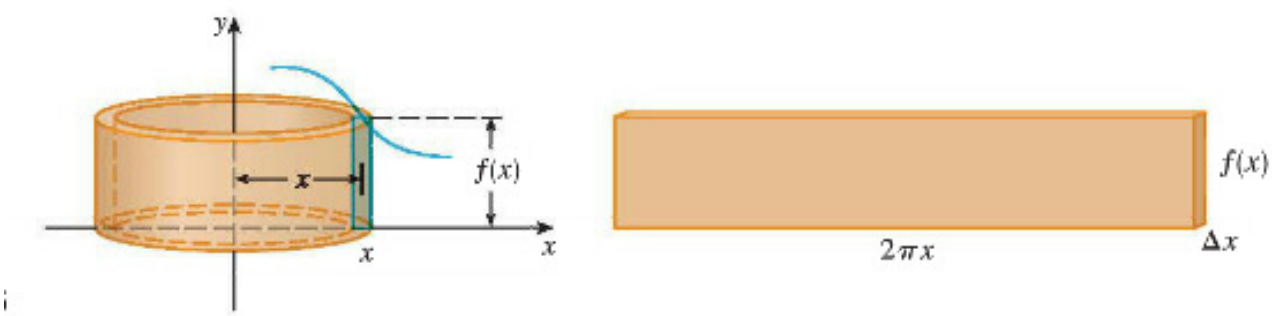
6.3 Volumes by Cylindrical Shells

The idea here is still to use an integral to add up small pieces, but each piece has a different shape in this section.







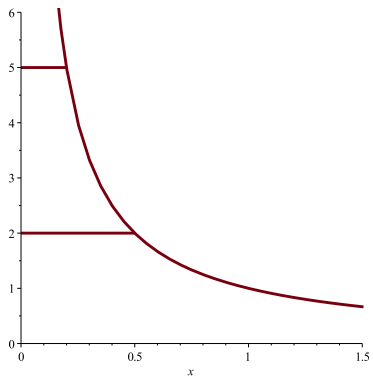
We use **paraLLel** slices to the axis of rotation for **sheLLs**.

1. Find the volumes of the solids found by rotating the regions bounded by the given curves around the specified axis/line. Use the shell method.

   (a) $y = 3x - x^2$, $y = 0$ about the y-axis

(b) $y = -x^2 + 8x - 7$, $y = -x + 11$ about the line $x = -3$

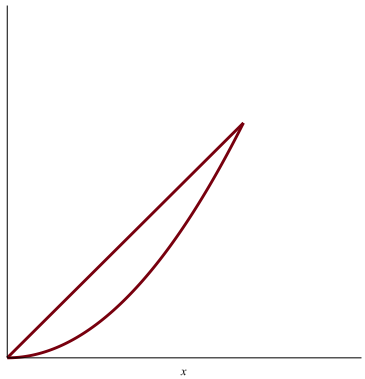
(c) $y = \frac{1}{x}$, $x = 0$, $y = 2, y = 5$ about the x-axis



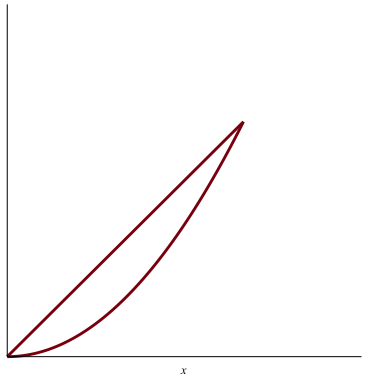
(d) $x = -(y-5)^2 + 4$, $x = 3$ about the line $y = 8$

2. Find the volumes of the solid found by rotating the region bounded by the given curves around the specified line, twice. Use the shell method in part (a) and the disk/washer method in part (b).

   (a) $y = x^2$, $y = 2x$ about the line $y = -3$.



   (b) $y = x^2$, $y = 2x$ about the line $y = -3$.



How to determine which method to use for solids of revolution if you aren't told which method to use:

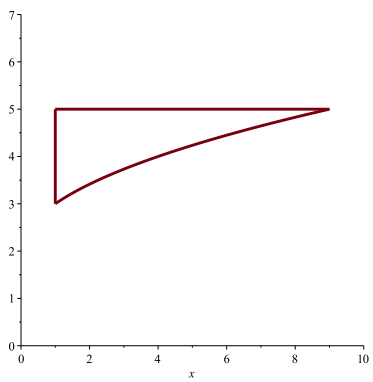
(i) If the method of slicing chosen above requires that you split up the integral into two pieces, does slicing the other way only require one integral? If the other way only requires one integral, is it easy to rewrite the equations in terms of the other variable?

(ii) If the direction of slicing does not matter, are your equations written in terms of x or y? If x, try a dx integral, which means vertical slices. If y, try a dy integral, which means horizontal slices.

(iii) If slices are **paraLLel** to axis of rotation, use **sheLLs**. If slices are perpendicular to axis of rotation, use disks/washers.

3. Find the volumes of the solids found by rotating the regions bounded by the given curves around the specified axis/line. Use any method.

   (a) $y = \sqrt{x} + 2$, $x = 1$, $y = 5$ about the line $x = 11$



(b) $y = x^2 - 6x$, $y = 4x - x^2$ rotated around $y = 9$.

(c) $y = 2x$, $y = 6 - x$, and the x-axis rotated around $y = 6$.

6.4 Work

Units are important in this section!

	Force	Distance	Work
SI Units	Newtons (N) To convert from mass in kg to Newtons, multiply by gravity: 9.8 m/s^2	Meters (m)	Joules (J) 1 N m = 1 J
Imperial Units	Pounds (lb)	Feet (ft)	Foot-Pounds (ft-lbs)

If a constant force F moves an object d units, then the work done is $W = Fd$.

1. How much work is done in lifting a 40 kg object to a height of 5 m?

$W = 40(9.8)(5) = 200(9.8) = 1960\,J$

What happens when force is not constant? Suppose an object is moved from $x = a$ to $x = b$ by a force $f(x)$ that varies depending on where the object is. Then, the work done by the force in moving the object from a to b is given by

$$W = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x = \int_a^b f(x)\,dx$$

2. When a particle is x ft from the origin, a force of $f(x) = \dfrac{1}{\sqrt{x+4}}$ pounds acts on the object. How much work is done in moving the object from $x = 1$ to $x = 8$ ft?

$W_i = \dfrac{1}{\sqrt{x+4}}\,dx$ $\quad W = \int_1^8 \dfrac{1}{\sqrt{x+4}} = 2(x+4)^{1/2}\Big|_1^8 = 2\sqrt{12} - 2\sqrt{5}$

Spring Problems

Hooke's Law: The force required to hold a spring x units stretched *beyond its natural length* is proportional to x. In other words, $f(x) = kx$ for some positive constant k, which we call the spring constant.

3. A spring has a natural length of 20cm. A force of 60 N is required to hold the spring stretched to a length of 50cm. How much work is done in stretching the spring from a length of 30cm to a length of 90cm?

$f(x) = k(x)$ $\quad .3 \to \frac{3}{10}$

$60 = k(.3)$

$200 = k$

$f(x) = 200x \to k$

$W = \int_{.1}^{.7} 200(x) = 100x^2\Big|_{.1}^{.7}$

$= 100(.49 - .01) = 48\,J$

4. Suppose the work needed to stretch a spring from its natural length to a length of 5 ft beyond its natural length is 30 ft-lb.

   (a) How much work is done in stretching the spring from 3 ft beyond its natural length to 120 in beyond its natural length?

$$W = \int_0^5 kx = 30$$

$$\frac{kx^2}{2}\Big|_0^5 = 30$$

$$\frac{25}{2}k = 30 \Rightarrow k = \frac{60}{25} = \frac{12}{5}$$

$$W = \frac{12}{5}\int_3^{10} x \Rightarrow \frac{6}{5}x^2\Big|_3^{10}$$

$$\frac{6}{5}(100 - 9) = \frac{6}{5}(91) \text{ ft-lbs}$$

   (b) How far beyond its natural length will a force of 60 lb keep the spring stretched?

$$f(x) = kx$$

$$60 = \frac{12}{5}x \Rightarrow \frac{300}{12} = x \Rightarrow \underline{\underline{25}}$$

Rope Pulling Problems

5. A 40 ft rope that weighs 120 lbs hangs off a 100 ft cliff. How much work is needed to pull the whole rope to the top of the cliff?

$$f(x) = 120 - 3x$$

x = rope pulled

$f(x)$ = weight remaining

$$W = \int_0^{40} 120 - 3x$$

d = liner rope density $= \frac{120}{40} = 3$ lb/ft

6. A rope that weighs 4 lb/ft is being used to pull a 50 lb bucket up a 70 ft well. How much work is done in lifting the bucket up the first 30 ft?

$f(x)$ = total weight − rope density

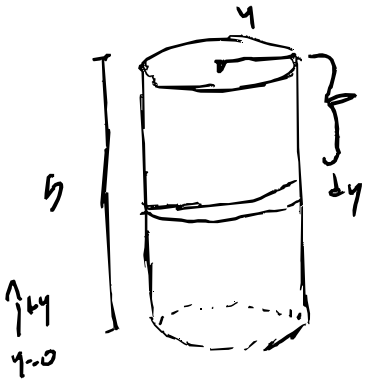
$= 4(70) + 50 - 4$

$= 330 - 4x$

$$W = \int_0^{30} 330 - 4x \, dx$$

Water Pumping Problems

7. A cylindrical tank has a radius of 4 m. The tank is 5 m tall and is filled with water. How much work is required to pump all the water out of the tank? Use the fact that the density of water is $\rho = 1000$ kg/m^3.



$d_i = 5 - y$

$A = \pi r^2$
$= \pi(16)$

$W = \int_0^5 9800(16\pi)(5-y)\,dy$

$W = 9800(16\pi)(5-y)$

$W = \rho g \int_6^7 (7-y)$

if $V = A_i \cdot d_i$

$F_i = m \cdot a_i$

$m_i = \rho_i \cdot V$

$V = A \cdot dy \Rightarrow A_i = \pi 16$

$F_i = 1000(16\pi)dy \cdot g\,dy$

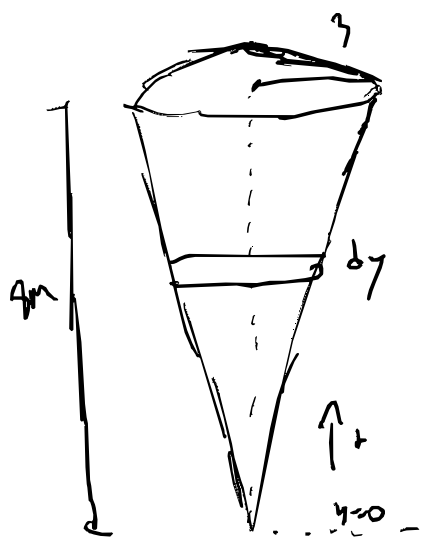
$m_i = 1000(16\pi)dy$

$V = 16\pi\,dy$

In general:

$$W = \rho g \int_a^b A_i d_i\,dy$$

8. An inverted conical tank is filled with water. The tank has a height of 8m and a radius of 3m. How much work is required to pump all the water out of the tank? Use the fact that the density of water is $\rho = 1000$ kg/m^3.



$W = \rho g \int_a^b A_i d_i\,dy$

$= \rho g \int_0^8 \frac{9\pi}{64} y^2 (8-y)\,dy$

$\frac{3}{8} = \frac{r}{y}$

$r = \frac{3}{8}y$

$A = \pi\left(\frac{3}{8}y\right)^2$

9. A trough is 10 ft long and has ends which are isosceles triangles with height 5 ft and width 4 ft. The trough has a spout at the top of the tank with height 1 ft. The trough is filled with water to a depth of 3 ft. How much work is required to pump all of the water out of the tank? Use the fact that the density of water is $\rho g = 62.5$ lb/ft^3.

10. A spherical tank with radius 6 m is half-filled with a liquid that has a density of $\rho = 200$ kg/m^3. The tank has a 2 m spout at the top. Find the work needed to pump all of the liquid out of the tank.

11. (If time) A trough is 20 m long with semicircular ends that have radius 6 m. If the tank is filled with water to a depth of 5 m, set up the integral to calculate how much work is needed to pump all the water out of the tank. Use the fact that the density of water is $\rho = 1000$ kg/m^3.